

FAKE NEWS DETECTOR

Elevate Labs Internship – Project Phase

Project Domain: Python Programming

Name: SARIPALLE SURESH

Tools: Python | Pandas | Scikit-learn | TF-IDF | Linear SVM | Streamlit | Google Colab | GitHub

GitHub: <https://github.com/sureshsaripalle2004/Fake-News-Detection>

1. INTRODUCTION

The rapid growth of online news and social media has made it difficult to distinguish between genuine and misleading information. Fake news can spread quickly and influence public opinion. Therefore, an automated system that can analyze news content and classify it as fake or real can be useful for preliminary news screening. The **Fake News Detector** project uses Natural Language Processing (NLP) and Machine Learning techniques to classify news articles into **FAKE** and **REAL** categories. The system processes the news text, converts it into numerical features using **TF-IDF**, and applies machine learning algorithms for classification. A Streamlit-based interface is also developed to allow users to enter news articles and obtain predictions with confidence scores.

2. ABSTRACT

The **Fake News Detector** is a machine-learning-based text classification project designed to identify whether a given news article is fake or real. The project uses the Fake.csv and True.csv datasets containing real and fake news articles. The datasets are combined, duplicate records are removed, and the news content is cleaned through text preprocessing. **TF-IDF (Term Frequency–Inverse Document Frequency)** is used to transform textual information into numerical feature vectors. Three machine learning algorithms—**Logistic Regression, Linear SVM, and Multinomial Naive Bayes**—are trained and evaluated. Among the evaluated models, **Linear SVM achieved the best performance with 99.77% accuracy and 99.75% F1-score**, and was therefore selected as the final classifier. Since Linear SVM does not directly provide probability estimates, calibration is applied to generate confidence scores. The final system is deployed through a **Streamlit application**, supporting both individual news classification and CSV-based batch prediction.

3. TOOLS USED

Tool / Technology	Purpose
Python	Overall project development
Pandas	Dataset loading and data manipulation
NumPy	Numerical operations
Scikit-learn	Machine learning and evaluation
TF-IDF Vectorizer	Text feature extraction
Logistic Regression	Classification model
Linear SVM	Final classification model
Multinomial Naive Bayes	Classification model
Matplotlib	Performance visualization
Joblib	Saving and loading the trained model
Streamlit	Web application development
Google Colab	Model development and experimentation
Cloudflare Tunnel	Temporary public access for demonstration

4. STEPS INVOLVED IN BUILDING THE PROJECT

Step 1: Dataset Collection

The Fake.csv and True.csv datasets were loaded and assigned class labels, where 0 represents Fake and 1 represents Real.

Step 2: Data Preparation

The datasets were combined into a single dataset. The title and article text were combined into a single content field. Duplicate records were removed and missing values were checked.

Step 3: Text Preprocessing

The news content was converted to lowercase. URLs, HTML tags, numbers, punctuation, and unnecessary whitespace were removed to prepare the text for feature extraction.

Step 4: Train-Test Split

The dataset was divided into training and testing sets using an 80:20 split while maintaining the class distribution.

Step 5: TF-IDF Feature Extraction

TF-IDF was applied to convert the cleaned news text into numerical feature vectors.

Unigrams and bigrams were used to capture individual words as well as word combinations.

Step 6: Machine Learning Model Training

Three classification algorithms were trained:

- Logistic Regression
- Linear SVM
- Multinomial Naive Bayes

Step 7: Model Evaluation and Selection

The models were evaluated using accuracy, precision, recall, and F1-score. Linear SVM achieved the highest overall performance:

- Accuracy: 99.77%
- Precision: 99.67%
- Recall: 99.83%
- F1-Score: 99.75%

Therefore, Linear SVM was selected as the final model.

Step 8: Confidence Calibration

Because Linear SVM does not directly provide probability estimates, calibration was applied to generate confidence scores for the application.

Step 9: Application Development

A Streamlit application was developed where users can enter a news headline or article and receive a FAKE/REAL prediction with confidence. The application also supports CSV upload for batch classification and allows users to download the prediction results.

Step 10: Testing and Deployment

The saved model was loaded and tested successfully. The Streamlit application was then made accessible for demonstration using a Cloudflare Quick Tunnel.

5. CONCLUSION

The **Fake News Detector** successfully demonstrates the application of Natural Language Processing and Machine Learning for fake news classification. The project combines data preprocessing, TF-IDF feature extraction, machine learning model comparison, performance evaluation, confidence estimation, and Streamlit deployment into a complete working application. Among the tested algorithms, **Linear SVM provided the best classification performance with 99.77% accuracy and 99.75% F1-score**. The final application provides both individual and batch news classification capabilities, making the project practical for preliminary screening of news content. However, the system is a **machine-learning-based classification tool and not a definitive fact-checking system**. Important news and claims should still be verified using reliable and independent sources.